Logic Gates

Contemporary Digital Electronic Systems — Chapter 4

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Why This Chapter Matters

Logic gates are the atoms of digital hardware. Every combinational circuit later in the module (adders, decoders, multiplexers, controllers) is wired from a handful of gate types. We need their symbols, truth tables and real electrical behaviour first.

After this chapter, you can:

1. Draw the symbol and truth table of AND, OR, NOT, NAND, NOR, XOR, XNOR.
2. Explain why NAND and NOR are universal; build a NOT and an AND from NAND.
3. Interpret propagation delay and power dissipation.
4. Compare CMOS and TTL power.
5. Recognise a tri-state buffer and where it belongs.

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An Opening Question

If you had a single gate type to build every digital circuit, which would you pick?

The answer may surprise you. We return to it with NAND.

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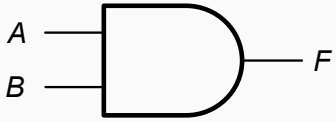
Gates, Boolean Algebra and Truth Tables



- A gate has one or more inputs and one output set by its function.
- Boolean algebra: variables take values 0 or 1; complement written $\overline{A}$ or A'.
- A truth table lists output for every input combination; inputs left, output right.
- A canonical truth table for n inputs has 2^n rows.

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AND Gate



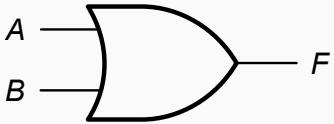
$$F = A \cdot B = AB$$

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

HIGH only when all inputs are HIGH. Like switches in series: the lamp lights only if both are closed.

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OR Gate



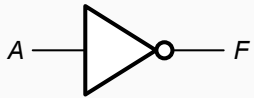
$$F = A + B$$

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

HIGH when at least one input is HIGH.
Like switches in parallel.

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NOT Gate (Inverter)



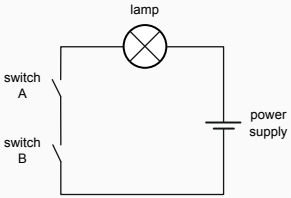
$F = \overline{A}$

A	F
0	1
1	0

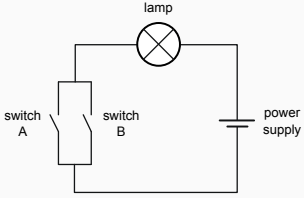
One input. Output is the opposite level. The bubble at the output denotes inversion.

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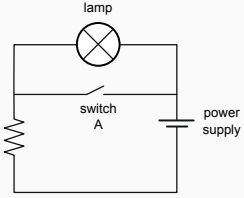
Switch Models of the Basic Gates



AND: series



OR: parallel

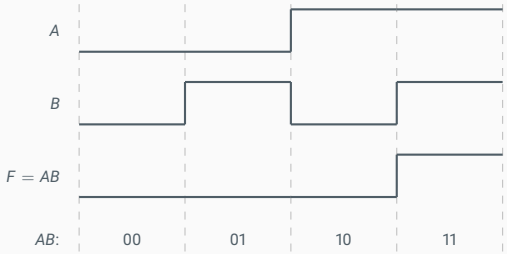


NOT: switch across the lamp

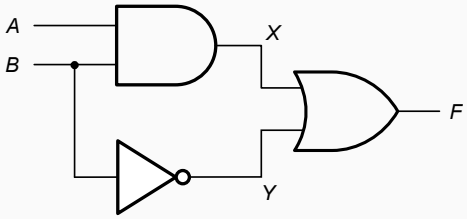
- Open switch = logic 0; closed switch = logic 1; lamp = output.
- This is why digital circuits are called switching circuits.

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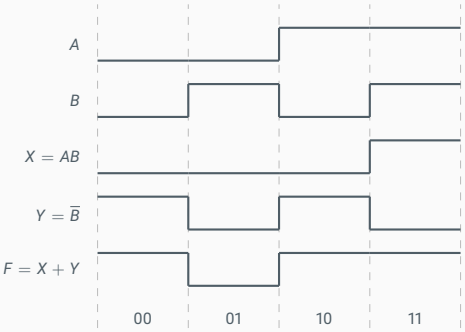
Timing Diagram of a Gate



- A truth table has no notion of time. A timing diagram plots levels against time.
- Each interval matches one truth-table row.
- Inputs on top, outputs below.



$X = AB, \quad Y = \overline{B}, \quad F = X + Y$



F is LOW only when X and Y are both LOW: $A = 0,\ B = 1$. For $AB = 10$: $X = 0$, $Y = 1$, so $F = 1$.

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More Than Two Inputs



- NOT has exactly one input; AND and OR can have many.
- AND is 1 only if every input is 1; OR is 1 if any input is 1.
- A 3-input gate has $2^3 = 8$ truth-table rows (000 to 111). The 3-input AND is 1 only on row 111.
- *n*-input NAND is 0 only when all inputs are 1; *n*-input NOR is 1 only when all inputs are 0.
- XOR is the exception: it is defined for two inputs. Parity over several bits is built by cascading two-input XOR gates (Chapter 9).

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NAND Gate



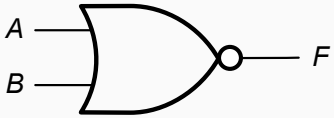
$$F = \overline{AB}$$

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

NOT-AND: LOW only when all inputs are HIGH. Equivalent to an AND followed by a NOT.

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NOR Gate



$$F = \overline{A + B}$$

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

NOT-OR: HIGH only when all inputs are LOW. Equivalent to an OR followed by a NOT.

NAND Is Universal

NOT from NAND

Tie both inputs together:
$\overline{A \cdot A} = \overline{A}$

AND from NAND

NAND followed by a NAND wired as NOT:
$\overline{\overline{AB}} = AB$

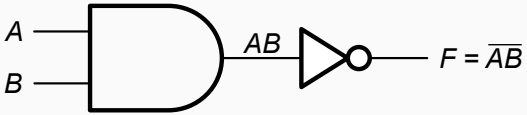
Universal gates

Any Boolean function can be built from NAND gates alone, or from NOR gates alone.

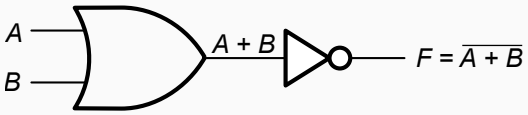
Back to the opening question: pick NAND (or NOR). AND alone or OR alone cannot invert, so they cannot build most functions. One gate type, one part number, every circuit (Section 7.7 shows a circuit rebuilt to fit one NAND package).

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Equivalent Forms of NAND and NOR

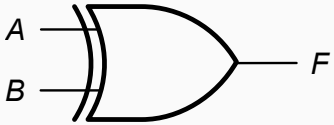


NAND = AND + NOT



NOR = OR + NOT

XOR Gate

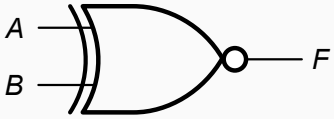


$$F = \overline{A}B + A\overline{B} = A \oplus B$$

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

HIGH when the inputs differ. Used in adders and in circuits that detect differences between binary values.

XNOR Gate



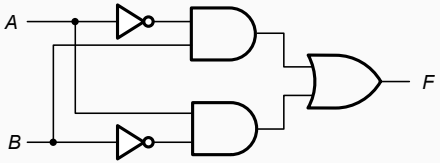
$$F = AB + \overline{A}\,\overline{B} = A \odot B$$

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

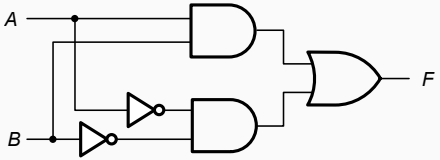
HIGH when the inputs are equal. Opposite of XOR. Useful for comparing bits: output is 1 when they match.

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XOR and XNOR from Basic Gates



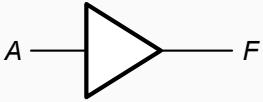
XOR from AND, OR, NOT



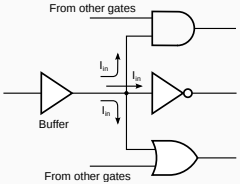
XNOR = XOR followed by NOT

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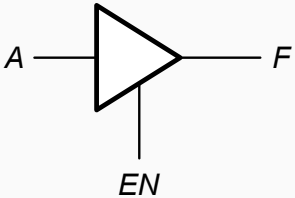
Buffers



- One input, one output; $F = A$. The logic value is unchanged.
- It supplies more current than the source, e.g. to drive an LED array, and isolates stages.
- Larger loads such as relays and solenoids need a proper driver circuit.

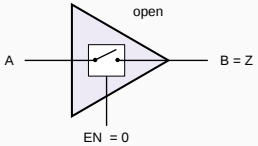


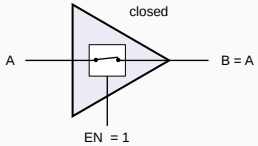
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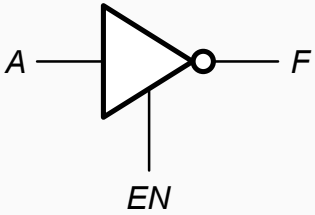
EN	*A*	*F*
1	0	0
1	1	1
0	X	Hi-Z

- An enable input controls the output.
- Enabled: output follows input. Disabled: high impedance (Hi-Z), electrically disconnected.
- Several devices can share one line (a bus) if only one drives it at a time.
- Variants: inverting, and active-low enable $\overline{EN}$.



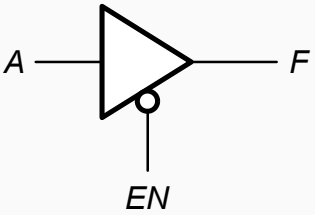


Buffer Variations



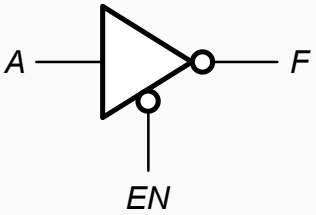
Inverting, active-high *EN*

EN	*A*	*F*
1	0	1
1	1	0
0	X	Hi-Z



Non-inverting, active-low $\overline{EN}$

$\overline{EN}$	*A*	*F*
0	0	0
0	1	1
1	X	Hi-Z



Inverting, active-low $\overline{EN}$

$\overline{EN}$	*A*	*F*
0	0	1
0	1	0
1	X	Hi-Z

Each gives signal isolation and controlled drive in a different configuration.

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- George Boole (1854) based it on AND, OR, NOT. Claude Shannon (1930s) applied it to two-state switching circuits.
- Theorems simplify expressions, cutting gates, cost and failure points.

$$
\begin{array}{cc}
0 \cdot 0 = 0 & 1 + 1 = 1 \\
1 \cdot 1 = 1 & 0 + 0 = 0 \\
0 \cdot 1 = 1 \cdot 0 = 0 & 0 + 1 = 1 + 0 = 1 \\
\overline{1} = 0 & \overline{0} = 1
\end{array}
$$

Here $+$ is OR, $\cdot$ is AND, and the overbar is NOT.

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	Theorem	AND form	OR form
T1	Annulment	$A \cdot 0 = 0$	$A + 1 = 1$
T2	Identity	$A \cdot 1 = A$	$A + 0 = A$
T3	Idempotency	$A \cdot A = A$	$A + A = A$
T4	Complements	$A \cdot \overline{A} = 0$	$A + \overline{A} = 1$
T5	Involution	$\overline{\overline{A}} = A$	

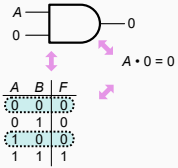
- Duality: swap AND with OR, and 0 with 1, to get the dual theorem.

Example: $F = A \cdot A \cdot \overline{A} + 1$

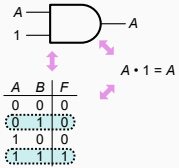
T3: $A \cdot A = A$. T4: $A \cdot \overline{A} = 0$. So $F = 0 + 1 = 1$. (Directly by T1: anything $+ 1 = 1$.)

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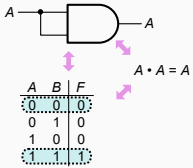
Theorems as Circuits: T1–T3



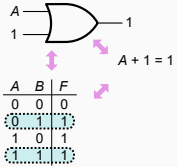
$A \cdot 0 = 0$



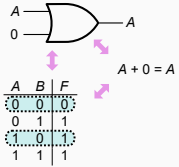
$A \cdot 1 = A$



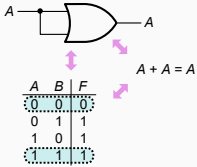
$A \cdot A = A$



$A + 1 = 1$



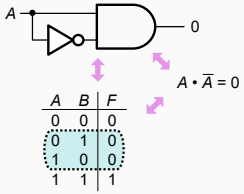
$A + 0 = A$



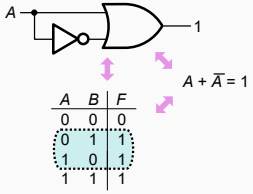
$A + A = A$

T1: one input forces the output on its own. T2: a 1 on an AND (0 on an OR) makes the gate transparent. T3: *A* can only equal itself, so the gate can be deleted.

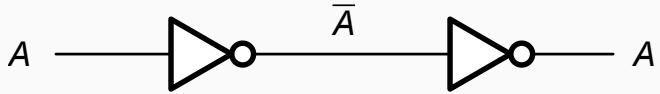
Theorems as Circuits: T4–T5



$A \cdot \overline{A} = 0$



$A + \overline{A} = 1$



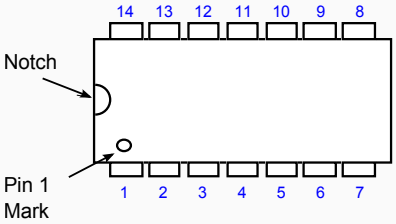
$\overline{\overline{A}} = A$

T4: A and $\overline{A}$ are never high together, and one is always high. T5: two inversions cancel, so the pair can be removed.

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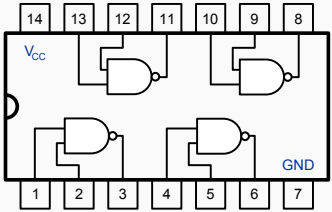
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- SSI chips of the 74 series hold four or six gates each: “quad” and “hex”.
- DIP suits labs and breadboards; SOT-23 and DFN fit tight spaces.
- DIP14: pins numbered counter-clockwise from pin 1, beside the notch.
- Pinouts vary: always check the datasheet.

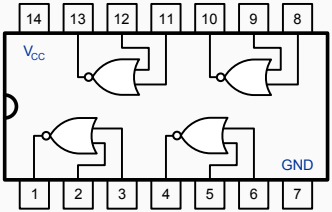


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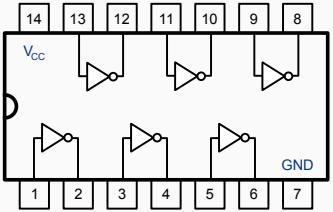
74-Series Examples



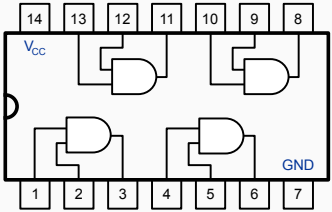
74x00 quad NAND



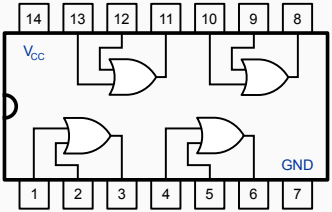
74x02 quad NOR



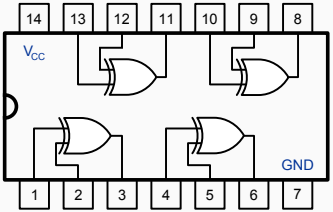
74x04 hex inverter



74x08 quad AND



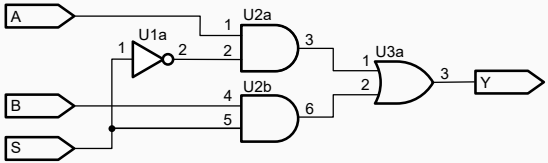
74x32 quad OR



74x86 quad XOR

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Reading a Schematic



- U1 (74x04): inverter U1a, pins 1–2.
- U2 (74x08): AND gates U2a (pins 1, 2, 3) and U2b (pins 4, 5, 6).
- U3 (74x32): OR gate U3a, pins 1, 2, 3.
- The designator names the IC; the letter names the gate inside it.

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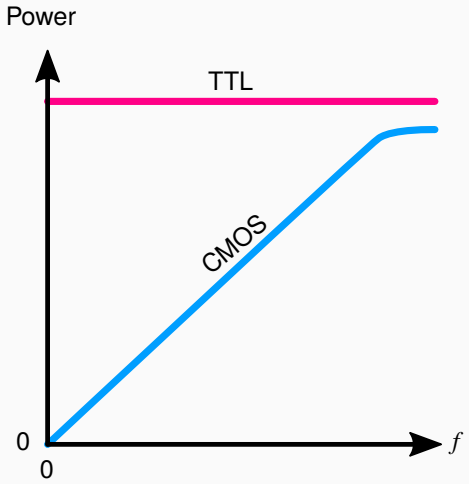
7 Alternative Symbols

8 Summary

CMOS total power = static + dynamic

$$P_{\mathrm{s}} = I_{\mathrm{dd}} V_{\mathrm{dd}} \quad \text{(leakage)}$$
$$P_{\mathrm{dyn}} = C_L V_{\mathrm{dd}}^2 f$$

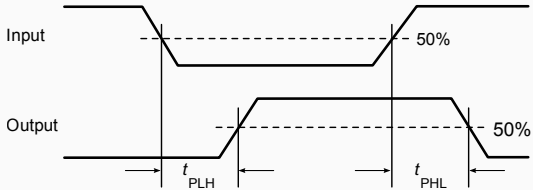
- C_L: load capacitance; V_{dd}: supply; f: switching frequency.
- CMOS idles at very low power; its power rises with f.
- TTL draws roughly constant power regardless of speed.



Propagation Delay

- A gate does not respond instantly. Propagation delay is the time from an input transition to the output transition.
- t_{PHL}: output HIGH to LOW. t_{PLH}: output LOW to HIGH.

$$t_{\mathrm{pd}} = \frac{t_{\mathrm{PHL}} + t_{\mathrm{PLH}}}{2}$$



- Typical speed order: inverters, then NAND/NOR, then AND/OR, then XOR/XNOR.
- The path with the greatest total delay is the critical path; it limits the maximum operating speed.

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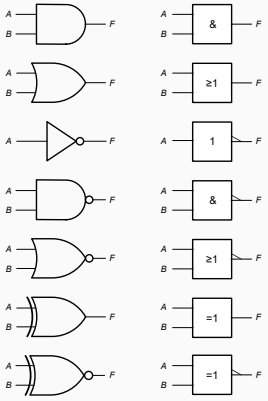
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Conventional and Rectangular Symbols



- ANSI/IEEE Std 91-1984 rectangular symbols: `&` = AND, `1` = OR, `=1` = exactly one input high (XOR for two inputs; XOR itself is defined for two). A triangle replaces the bubble.

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Summary

- A truth table defines a gate's function; the symbol is its graphical form.
- NAND and NOR are universal: any Boolean function from one gate type.
- Propagation delay limits speed; the critical path sets the maximum frequency.
- Dynamic CMOS power $\approx CV_{\mathrm{DD}}^2 f$ for constant switching activity.
- A tri-state buffer adds a Hi-Z state so outputs can share a bus, one driver at a time.

Looking ahead

Chapter 5 uses Boolean algebra to simplify the logic that these gates implement.